APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Scheme for Valuation/Answer Key

Scheme of evaluation (marks in brackets) and answers of problems/key

SIXTH SEMESTER B.TECH DEGREE(S) EXAMINATION MAY 2023(2019 Scheme)

Course Code: CST304

Course Name: COMPUTER GRAPHICS AND IMAGE PROCESSING

Max. Marks: 100 | Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks. — Marks

1 — Each application 0.5 marks; total 6x0.5 = **3 marks** — (3)

2 — (3)

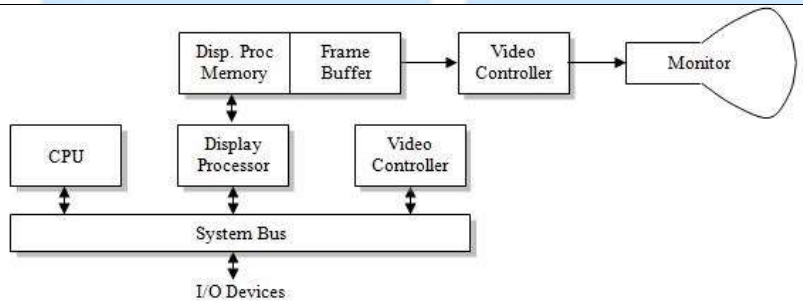

Diagram only: **2 marks**; diagram with brief expln: **3 marks**

3 — (3)

Food Fill Algorithm	Boundary Fill Algorithm
1.In flood fill algorithm the area is defined with multiple colour	1 In Boundary fill algorithm the area is defined with single colour.
2. The interior is coloured with any colour.	2. The interior point is replaced with new colour.
3. The old colour is replaced with new colour.	3. The interior points are replaced with new colour.
4. A flood fill may use an unpredictable amount of memory to finish because it isn't known how many sub-fills will be spawned.	4. Boundary fill is usually more complicated but it is a linear algorithm and doesn't require recursion.
5. It is Time Consuming.	5. It is less time Consuming.

Each difference: **1 mark**; any 3 difference: **3 marks**

4 — (3)

Homogeneous coordinates are a coordinate system with wide applications in geometry and computer graphics.
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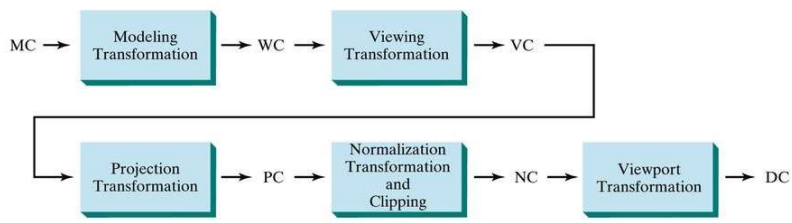
Homogenous coordinate definition: **1.5 marks**
Need for homogenous coordinate: **1.5 marks**

5		$xv = Sx.xw + tx \quad (1)$ $yv = Sy.yw + ty \quad (2)$ $\begin{bmatrix} xv \\ yv \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & tx \\ 0 & 1 & ty \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} Sx & 0 & 0 \\ 0 & Sy & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} xw \\ yw \\ 1 \end{bmatrix}$ where $Sx = \frac{(xv_{max} - xv_{min})}{(xw_{max} - xw_{min})}, Sy = \frac{(yv_{max} - yv_{min})}{(yw_{max} - yw_{min})}$ $tx = \frac{xw_{max}v_{min} - xw_{min}xv_{max}}{(xw_{max} - xw_{min})}, ty = \frac{yw_{max}yv_{min} - yw_{min}yv_{max}}{(yw_{max} - yw_{min})}$	(3)
6			(3)
7		Let *S* represent a subset of pixels in an image. Two pixels *p* and *q* are said to be *connected in S* if there exists a path between them consisting entirely of pixels in *S*. For any pixel *p* in *S*, the set of pixels that are connected to it in *S* is called a *connected component* of *S*.	(3)
8		Sampling : **1.5 marks** Quantization : **1.5 marks**	(3)
9		Very brief description: **1 mark**; detailed description with filter equation or kernel: **3 marks**	(3)
10		Suppose an image, $f(x, y)$, is composed of light objects on a dark background, in such a way that object and background pixels have intensity values grouped into two dominant modes. One obvious way to extract the objects from the background is to select a threshold, *T*, that separates these modes. Then, any point (x, y) in the image at which $f(x, y) > T$ is called an *object point*. Otherwise, the point is called a *background* point. In other words, the segmented image, denoted by $g(x, y)$, is given by $g(x,y) = \begin{cases} 1 & \text{if } f(x,y) > T \\ 0 & \text{if } f(x,y) \le T \end{cases}$ When *T* is a constant applicable over an entire image, the process given in this equation is referred to as *global thresholding*. When the value of *T* changes over an image, we use the term *variable thresholding*.	(3)
		PART B	
		Answer one full question from each module,eachcarries 14 marks.	
		Module I	
11	a)	p_0 =2dy-dx (Initial decision parameter) : derivation **1 mark** If $p_k<0$, $p_{k+1} = p_k+2dx$: derivation **2 marks** If pk>0, $p_{k+1} = p_k+2(dx-dy)$: derivation **2 marks**	(7)

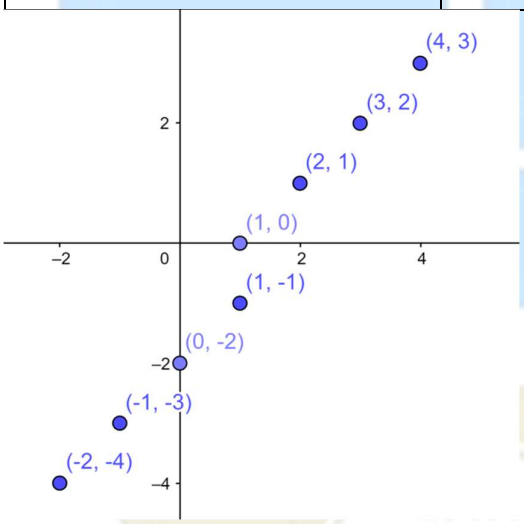
diagrams: **2 marks**

b) Slope (m) = (3-(-4))/(4-(-2)) = 7/6 = 1.16 >1 (7)

$y_{k+1}=y_k+1$; $x_{k+1}=x_k+1/m$

Initial (x_k,y_k) => (-2,-4)

x_{k+1}	y_{k+1}
-2+0.86 = -1.14 (-1)	-4+1=-3 (-3)
-1.14+0.86 = -0.28 (0)	-3+1=-2 (-2)
-0.28+0.86=0.58 (1)	-2+1=-1 (-1)
0.58+0.86 = 1.44 (1)	-1+1=0 (0)
1.44+0.86 = 2.3 (2)	0+1=1 (1)
2.3+0.86 = 3.16 (3)	1+1=2 (2)
3.16+0.86 = 4.02 (4)	2+1=3 (3)



Finding slope: 1 mark

Updation equations : 2 marks

Pixel coordinates calculated (end points inclusive): 8 x 0.5 marks =**4 marks**

OR

12 a) **Out of Syllabus – credit to be given if attempted.** (7)

b) Centre (h,k) => (5,3) (7)

radius r =5

If d_i>=0 then $x_{i+1}=x_i+1$; $y_{i+1}=y_i-1$; $d_{i+1}=d_i+4(x_i-y_i)+10$

If d_i<0 then $x_{i+1}=x_i+1$; $y_{i+1}=y_i$; $d_{i+1}=d_i+4x_i+6$

Initial d_0=3-2r = 3-2x5 = -7

calculation of initial decision parameter **(1 mark)**

Let take centre at origin. Starting point (0,r) => (0,5)

Step	x_i	y_i	x_{i+1}	y_{i+1}	d
1	0	5			3-2r = -7 < 0
2	0	5	1	5	d_{i+1}=-7+4(0)+6 = -1 < 0
3	1	5	2	5	d_{i+1}=-1+4(1)+6 = 9 > 0
4	2	5	3	4	d_{i+1}=9+4(2-5)+10 = 7 > 0
5	3	4	4	3	x >= y stop

Finding coordinates in only first quadrant each pair 1 mark **(3 marks)**

decision parameter correct usage : **(2 marks)**

Actual circle in first quadrant after adding the centre (5,3) as offset

x_i+h	y_i+k
0+5=5	5+3=8
1+5=6	5+3=8
2+5=7	5+3=8
3+5=8	4+3=7

Find the remaining coordinates using 8-way symmetricity **(1 mark)**

Module II

13 a) (i) Rotate the square by 45° about its vertex (2,6) (8)

Step 1: Translate the square such that its vertex (2,6) coincides with the origin

tx=-2, ty=-6

Step 2: Rotate by 45° anti-clockwise

Step 3: Translate the square back by tx=2, ty=6

[Steps are correct and matrices or equations used are correct : 4 marks; Steps not followed direct use of matrix or equations: 2 marks]

(ii)

Step 1: Translate the square such that its centre (4,4) coincides with the origin

tx=-4, ty=-4

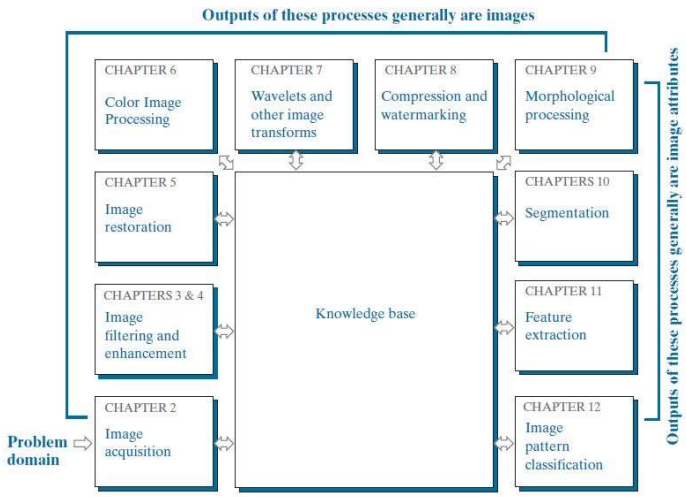
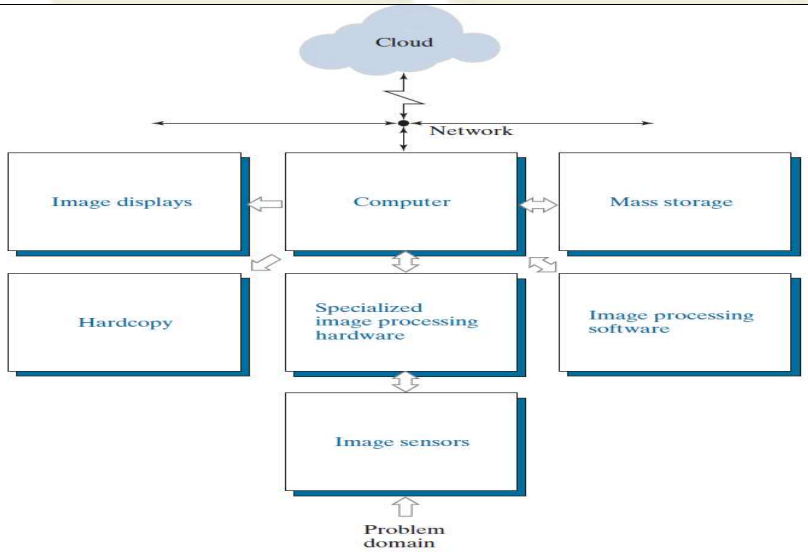
Step 2: Scale the square by using Sx=Sy=2

Step 3: Translate the square back by tx=4, ty=4

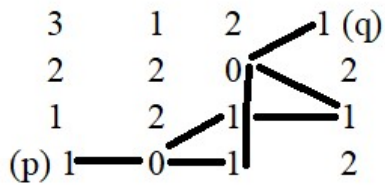
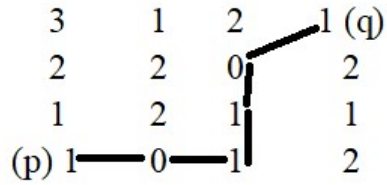
[Steps are correct and matrices or equations used are correct : 4 marks; Steps not followed direct use of matrix or equations: 2 marks]

b) 3D transformations – translation, scaling, rotation using homogenous (6)

		coordinates: 2 marks each; total 3 x 2 marks = **6 marks**	
		OR	
14	a)	scan line polygon filling algorithm : **5 marks** data structures used : **3 marks**	(8)
	b)	(i) two successive 2D rotations are additive Show that $R(\theta_1)R(\theta_2) = R(\theta_1+\theta_2)$: **3 marks** (ii) two successive 2D scaling are multiplicative. Show that $S(s_{x2},s_{y2})S(s_{x1},s_{y1}) = S(s_{x1}.s_{x2},s_{y1}.s_{y2})$: **3 marks**	(6)
		Module III	
15	a)	Let A(X0,Y0) => A(-1,-2) ; B(X1,Y1) => A(9,7) Find region outcode of end points A(0101) and B(1010) : **1 mark** Perform OR operation 0101 OR 1010 = 1111 (15) Perform AND operation 0101 AND 1010 = 0000 (0) => Line segment is partially visible needs to be clipped Starting from end point 'A' clip w.r.t to bottom boundary of clipping window X=X0+(X1-X0)*(Ymin-Y0)/(Y1-Y0) = -1+(9--1)*(0--2)/(7--2) = 1.22 Intersection with bottom boundary A'(X,Ymin) => A'(1.22,0) Intersection with top boundary B'(X,Ymax) **Algorithm steps : 2 marks; clipped result: 2 marks; intersection point calculation: 1 mark**	(6)
	b)	Parallel projections: Orthographic – (Front, Rear, Top) **[1 marks]** OR Multiview **[1 marks]** Axonometric – Isometric, Dimetric, Trimetric **[3 marks]** Oblique Cabinet, Cavalier **[2 marks]** Perspective One point,two point,three point **[2 marks]**	(8)
		OR	
16	a)	Sutherland-Hodgeman polygon clipping with respect to each boundary showing the steps : **2 marks**;	(8)

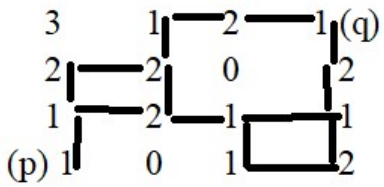
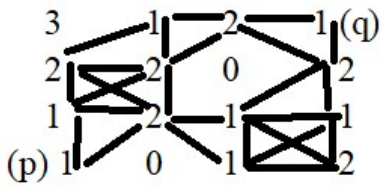
		4x2 marks= 8 marks Only clipped diagram at each stage shown : 4 **marks** Only final result shown : 3 **marks**	
	b)	Depth buffer algorithm : 4 **marks** example : 2 **marks**	(6)
		Module IV	
17	a)	 Diagram : **4 marks**; brief explanation of fundamental steps: **4 marks**	(8)
	b)	Spatial resolution: **3 marks** Gray level resolution : **3 marks**	(6)
		OR	
18	a)	 Diagram : 4 marks; brief explanation of components: 4 marks	(6)
	b)	V={0,1}	(8)

4-connected path not possible

	
Shortest 8-connected path length is 4	Shortest m-connected path length is 5

All three path lengths answered for V={0,1}: 4 **marks**

V={1,2}

	
Shortest 4-connected path length is 6 Shortest m-connected path length is 6	Shortest 8-connected path length is 4

All three path lengths answered for V={1,2}: 4 **marks**

Module V

19 a) i) Image Negatives: 3 **marks** (6)
ii) Log Transformation: 3 **marks**

b) (8)

rk	0	1	2	3	4	5	6	7
no. of pixels (nk)	0	1	2	2	1	2	6	2

total number of pixels (n)=16; frequency table : **2 marks**

r_k	n_k	$pr(r_k)=n_k/n$	$\sum_{j=0}^{k} p_r(r_j)$	$(L-1)\sum_{j=0}^{k} p_r(r_j)$	S_k	
0	0	0	0	0	0	
1	1	1/16=0.06	0.06	0.42	0	
2	2	2/16=0.12	0.18	1.26	1	

3	2	2/16=0.12	0.3	2.1	2			
4	1	1/16=0.06	0.36	2.52	3			
5	2	2/16=0.12	0.48	3.36	3			
6	6	6/16=0.37	0.85	5.95	6			
7	2	2/16=0.12	0.97	6.79	7			

Histogram equalization steps: **4 marks**

Plotting histogram before and after : **2 marks .**

		OR	
20	a)	i) Sobel : 4 **marks** ii) Prewitt: 4 **marks**	(8)
	b)	Region splitting : **3 marks** Region merging: **3 marks**	(6)

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